

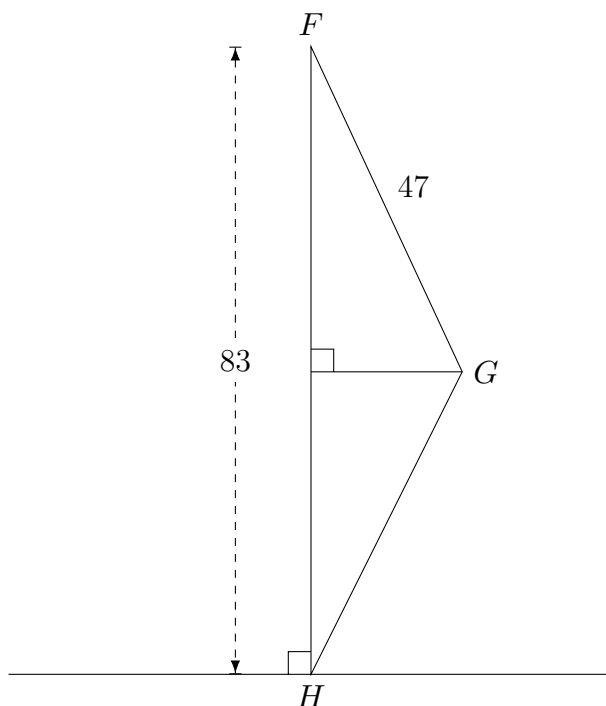
Standard 2 Trigonometry

Question Bank

Question 1

The triangle FGH is shown below. The length of FG is 47 and the length of FH is 83.

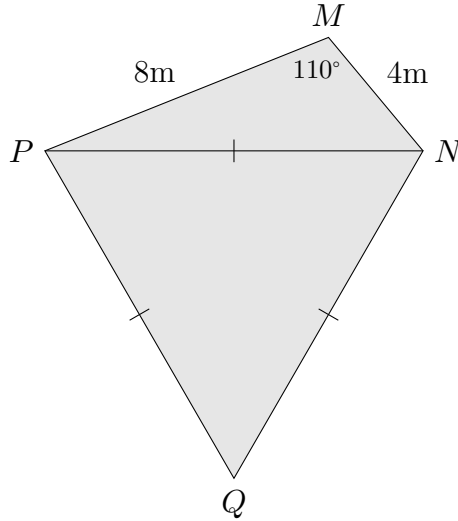
The angle of elevation of G from H is 64° and $\angle FGH$ is obtuse.



Find the length of GH correct to two decimal places.

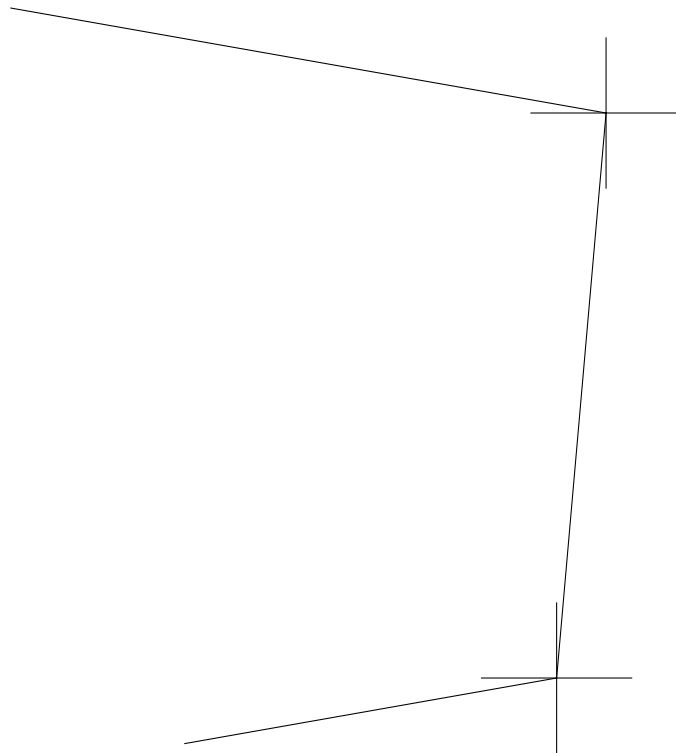
Question 2

A strange window is constructed from two triangles, PMN and PNQ . The length of PM is 8m and the length of MN is 4m. The $\angle PMN$ is 110° .



The window is to be coated in a protective film that costs $\$45/\text{m}^2$, how much will it cost to coat the window?

Question 3



Answers

Question 1

Since $\angle FHG$ is complementary to 64° ,

$$\angle FGH = 26^\circ.$$

Find $\angle FGH$ using sine rule,

$$\begin{aligned}\frac{\sin(\angle FGH)}{83} &= \frac{\sin(26^\circ)}{47} \\ \sin(\angle FGH) &= \frac{\sin(26^\circ)}{47} \times 83 \\ \angle FGH &\approx 50.7^\circ\end{aligned}$$

But since $\angle FGH$ is obtuse,

$$\angle FGH \approx 180^\circ - 50.7^\circ = 129.3^\circ$$

$$\angle GFH = 180^\circ - 129.3^\circ - 26^\circ = 24.7^\circ$$

$$\begin{aligned}\frac{GH}{\sin(24.7^\circ)} &= \frac{47}{\sin(26^\circ)} \\ GH &= \frac{47}{\sin(26^\circ)} \times \sin(24.7^\circ) \\ &\approx 44.80\end{aligned}$$

Question 2

The area of $\triangle PMN$ is

$$A_{\triangle PMN} = \frac{1}{2} \times 4 \times 8 \times \sin(110^\circ) \approx 15.04\text{m}^2$$

The length of PN

$$\begin{aligned}PN^2 &= 4^2 + 8^2 - 2 \times 4 \times 8 \times \cos(110^\circ) \\ PN &= \sqrt{101.889...} \\ &\approx 10.1\text{m}\end{aligned}$$

The area of equilateral $\triangle PNQ$

$$A_{\triangle PNQ} = \frac{1}{2} \times 10.1^2 \times \sin(60^\circ) \approx 44.17\text{m}^2$$

The cost is

$$(15.04 + 44.17) \times 45 = \$2664.45$$